Milestone 4 - Evaluation and presentation

Project: Stock Market Trend Analysis | **Week:** 13 | **Weight:** 7.5% of course | **Seed:** 42 **Notebook:** `notebooks/04_evaluation_presentation.ipynb`

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1. The journey: question to data to model to insight to limitations

The question was deliberately falsifiable: using ten years of daily prices for four tickers (^GSPC, AAPL, AMZN, NVDA), can a model forecast the next day's log return better than a naive baseline? We did not assume the answer was yes. Stakeholders are a retail investor, a portfolio analyst, and an academic reviewer, each of whom needs an honest number rather than a hopeful one.

The first consequential choice came in M1. An ADF test showed that price levels are non-stationary (p at least 0.86) while log returns are stationary (p = 0.0000), so we modeled log returns, not prices. Prices cannot be fit sensibly by ARIMA, and that single test set the target for the whole project. We pinned the download window and committed the raw CSV snapshots so the pipeline reproduces, split the data chronologically (train through 2023, validation 2024, holdout 2025 onward), and built the target as the next day's return per ticker so no label could leak backward across a split.

M2 told us what kind of signal exists. Returns have almost no linear autocorrelation (lag-1 near -0.16, which is microstructure, not a trading edge), but squared returns are strongly autocorrelated (lag-1 near +0.49). In plain terms, the direction of tomorrow is close to unpredictable, while the size of tomorrow's move clusters and is predictable. Mutual information confirmed that volatility features carry most of the usable signal, which is why M3 brought in a GARCH variance model and a nonlinear LSTM rather than only a linear one.

M3 built baselines, per-ticker ARIMA, a GJR-GARCH variance model, a LightGBM on 39 features, an attention LSTM, and an ensemble, and opened the holdout exactly once. The honest result: no model beats the naive forecast on error, and the two models that show a statistically significant directional edge turn out to predict up almost every day, so their accuracy is really the market's upward drift in a calm period.

We iterated, and some of it was uncomfortable. An off-by-one bug in the ARIMA one-step forecast had produced an apparent edge that was an artifact, and fixing it removed the edge. An un-tuned LightGBM overfit the holdout, so we replaced the single validation peek with a purged walk-forward cross-validation and Optuna, which shrank the model to two trees. We proposed five new volatility features and the ablation rejected all five. The LSTM looked promising at 54 percent on one run and then failed to reproduce that edge on the real GPU run. Each of these is in the decisions log, and together they are the point of the project: the efficient-market ceiling is real, and the interesting work was measuring exactly where it sits.

2. Final performance on the sealed holdout

Holdout period 2025-01-02 to 2026-05-19, 1,380 rows, opened once in M3. Naive_zero predicts a zero return every day. The perfect-foresight columns compare each model against an oracle (RMSE of 0, directional accuracy of 1.0) so the "how much of the gap did we close" question has an answer. Numbers reconcile with M3.md section 8 and are recomputed in `notebooks/04`.

Model	RMSE	MAE	DirAcc	DirAcc p (vs 0.50)	DM vs naive	frac. predicted up
naive_zero	0.02109	0.01382	n/a	-	-	-
ARIMA	0.02107	0.01379	0.514	0.294	0.783 (tie)	0.82
GJR-GARCH-t (mean)	0.02106	0.01375	**0.543**	**0.002**	0.589 (tie)	1.00
LightGBM (baseline)	0.02168	0.01431	0.483	0.226	**0.003 (worse)**	0.48
LightGBM (tuned)	0.02107	0.01377	**0.542**	**0.002**	0.533 (tie)	1.00
LSTM+Attention (real GPU)	0.02031*	0.01350	0.491	0.574	**0.004 (worse)**	0.51
Ensemble GARCH+LSTM	0.02005*	0.01318	0.512	0.424	0.823 (tie)	0.68

*Do not read the LSTM and ensemble RMSE as beating naive. They are scored on the 1,140 sequenced rows only (the first 60 days per ticker have no window), where the correct same-rows naive is **0.02004**, so both are actually worse than naive on their own rows (the DM column confirms it, p = 0.004 and 0.823). Every model's RMSE is compared against naive on its own rows, which is why `beats_naive` comes back empty.

What this says, read honestly:

1. No model beats naive on RMSE. The Diebold-Mariano test is not significant for any model that ties, and it is significant in the wrong direction for the un-tuned LightGBM and the LSTM, which are worse than naive. Gap closure against a perfect forecast is essentially zero.
2. Two models show a directional edge that is significant against a 0.50 coin flip, GJR-GARCH at 0.543 and the tuned LightGBM at 0.542, both at p = 0.002. That looks like skill until you check the last column: both predict up on 99 to 100 percent of days. The holdout up-day frequency is 0.542, which is exactly the tuned LightGBM's accuracy. The 0.50 null only detects "predicts up." Re-tested against the honest skill null, the up-day base rate of 0.542, the edge vanishes: GJR-GARCH gives binomial p = 0.978 and the tuned LightGBM p = 1.000. So the edge is the market's drift in a calm year, not day-to-day timing skill.
3. The attention LSTM's earlier 0.542 did not reproduce. The real GPU run lands at 0.491, a coin flip. The earlier figure was run-dependent noise, and reporting the reproduced number is the honest choice.

3. Error analysis

The ten worst forecasts for the promoted model (tuned LightGBM) are every large market move in the holdout, and the model predicts near zero for all of them. See `models/m4_top10_worst.csv`.

date	ticker	actual	predicted	abs error	what happened
2025-01-24	NVDA	-0.186	0.001	0.187	DeepSeek AI shock the next session

date	ticker	actual	predicted	abs error	what happened
2025-04-08	NVDA	0.172	0.002	0.170	tariff pause, sharp relief rally next session
2025-04-08	AAPL	0.143	0.002	0.141	same relief rally
2025-04-08	AMZN	0.113	0.001	0.112	same relief rally
2025-04-02	AAPL	-0.097	0.001	0.098	tariff announcement, risk-off crash
2025-04-02	AMZN	-0.094	0.001	0.095	same crash
2025-02-28	NVDA	-0.091	0.001	0.092	late-February AI drawdown
2025-10-30	AMZN	0.092	0.001	0.090	mega-cap earnings reaction
2025-02-26	NVDA	-0.089	0.001	0.090	large idiosyncratic move
2025-04-08	^GSPC	0.091	0.001	0.090	tariff pause, index-wide relief rally

The pattern is the whole story. Every one of these is news: a research shock, a tariff decision, an earnings report. None of it is visible in past prices, so a price-history model cannot see it coming and defaults to predicting roughly zero. That is not a bug, it is the efficient-market hypothesis showing up in the error table.

Three figures support this:

- `reports/figures/M4_fig_err_dist.png`: the absolute-error distribution is tiny on most days with a fat right tail of missed shocks (99th percentile far from the median).
- `reports/figures/M4_fig_err_by_month.png`: error spikes in April 2025, the tariff month, and is low the rest of the time. The model is worse exactly when the market is most volatile.
- `reports/figures/M4_fig_confusion.png`: the directional confusion matrix has an empty "predicted down" row. The tuned model never calls a down day. Its 0.542 accuracy is just the up-day rate.

4. Limitations

1. Efficient-market hypothesis. Public price history is already priced in, so predictability beyond chance is bounded for any model trained on prices alone. This is the binding constraint, and our own results confirm it.
2. Regime shift. The holdout (2025 onward) includes a tariff cycle and an AI cycle that the training window (2016 to 2023) could not contain. A model fit on one regime should not be trusted to generalize to another.
3. Black swans. The DeepSeek shock and the tariff crash are exactly the events the model misses. The forecasts carry no error bars for events like these.
4. Calm single holdout. The 2025 to 2026 holdout is lower volatility than the training period. Any conclusion about performance is a conclusion about one calm regime, not about a bear market or a crisis.
5. Survivorship bias. AAPL, AMZN, and NVDA are current large-cap survivors, and ^GSPC is an index that is not directly tradeable. A fair evaluation would include tickers that struggled or were delisted after 2016.
6. Daily granularity. Intraday signal such as opening gaps is invisible to a daily model.
7. Multiple comparisons. Six model variants were tested. The more models tried, the more likely one clears $p < 0.05$ by chance, so a single significant number should be read with that in mind. The two survivors

land at the same 0.542 to 0.543 because both predict up on almost every day, so both are capturing the same unconditional upward drift rather than an independent signal. Treat them as one weak effect, not two discoveries. (Note the lag-1 return autocorrelation the EDA found is negative, about -0.16, which is mean-reversion. A model exploiting it would predict down after up days and switch direction; these always-up models do the opposite, so the drift, not lag-1, is what they ride.)

5. Ethics and misuse audit

5a. Misuse surface

A retail investor who trades on these forecasts can lose real money. Directional accuracy for next-day stock returns sits in the 50 to 55 percent range, which is barely above a coin flip and below transaction costs for most retail brokerages. Our own backtest is the evidence: a timing rule that edges buy-and-hold at zero cost falls below it once a realistic 10 basis points per trade is applied. Presenting this model as a way to beat the market would be misleading and financially harmful.

5b. Required disclaimer

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5c. Fairness and equity

The dataset has no human subjects, so classical fairness metrics do not apply directly, but two equity points still matter. First, a model trained on US large-cap tech will not generalize to small-cap, emerging-market, or non-equity assets, so any claim of usefulness is narrow. Second, there is an asymmetry-of-information harm: a retail trader following this model competes against quant funds with far better data, lower latency, and more compute. A tool that encourages that trade transfers risk to the least-equipped participant.

5d. Data protection

Yahoo Finance data is public and contains no personal information. The picture changes with a real broker's data. Customer order flow is highly sensitive, and using it would require compliance with regulations such as GDPR or CCPA for personal data, and FINRA or MiFID II for market conduct, depending on the jurisdiction. None of that applies to this educational dataset, but it would apply to any production version.

6. Backtest narrative

The rule is simple: go long ^GSPC tomorrow when the model predicts a positive return, otherwise hold cash, over the 345-day holdout for ^GSPC. Two honest findings come out of it.

The promoted tuned LightGBM predicts up every day, so its signal never leaves the market and its backtest is identical to buy-and-hold (0 position switches). It captures the market's drift and adds no timing value. To show what happens to a model that does try to time, we use the ARIMA signal, which switches position 174 times.

strategy	terminal multiple	hypothetical $10,000
buy-and-hold ^GSPC (= promoted model)	x1.267	$12,666

strategy	terminal multiple	hypothetical $10,000
ARIMA timing, 0 bps cost	x1.274	$12,741
ARIMA timing, 10 bps cost	x1.071	$10,707

At zero cost the ARIMA timer edges buy-and-hold, and its Sharpe ratio is higher (1.18 versus 0.99, both computed in notebook 04 with a risk-free rate of 0 and cash days earning 0, which is conservative against the timer). Apply 10 basis points per trade across 174 switches and the terminal value drops below buy-and-hold, with the Sharpe ratio collapsing to 0.33. The daily strategy returns are not even significantly positive at zero cost (one-sample t-test t = 1.38, p = 0.168, computed in notebook 04). This is a hindsight backtest on one calm regime, not a tradeable strategy, and the method of evaluation, not the dollar figure, is what matters. See `reports/figures/M4_fig_backtest_cum.png`.

7. Reproducibility

The pipeline runs from the committed raw snapshots with a pinned date window and seed 42.

```
py -3.10 -m venv .venv && .venv\Scripts\activate
python -m pip install -r requirements.txt
python notebooks/01_data_collection_preprocessing.py   # raw snapshots -> features -> splits
python notebooks/02_eda.py                             # EDA on train only
python notebooks/03_model_building.py                  # models, holdout opened once
python notebooks/04_evaluation_presentation.py         # this milestone: tables, figures, backtest
```

The classical models, LightGBM, and all M4 outputs run on a local CPU. The 20-epoch attention LSTM needs a GPU, so it runs on Google Colab (see `reports/COLAB_TRAINING_GUIDE.md`); its predictions are already saved in `data/processed/holdout_predictions.parquet`. Every library version is pinned in `requirements.txt`. The root `README.md` carries the same steps.

8. Self-audit

Criterion	Status	Evidence
Journey narrative present (question to limitations)	done	section 1
Holdout numbers vs baselines reported	done	section 2 table, `models/m4_holdout_scores.csv`
Perfect-foresight gap-closure reported	done	section 2, notebook 04
Error analysis with top-10 worst + by-month plot	done	section 3, `M4_fig_err_by_month.png`, `m4_top10_worst.csv`
Directional confusion matrix	done	`M4_fig_confusion.png`

Criterion	Status	Evidence
At least 5 named finance-specific limitations	done	section 4 (7 listed)
Not-investment-advice disclaimer present	done	header, section 5b, slide 1
Backtest with explicit costs + disclaimer	done	section 6, `M4_fig_backtest_cum.png`
Reproducibility steps documented	done	section 7
Presentation deck with speaker notes	done	`reports/M4_presentation.html`
All four milestone reports present	done	`reports/milestones/M1.md` to `M4.md`

Notebook `04_evaluation_presentation.py` prints all 12 of its self-audit checks as passing, each bound to a computed result rather than a hardcoded value.

Honest closing

The headline is a negative result, and that is the finding. No model trained on price history beats a naive forecast of next-day returns out of sample. The one directional edge that clears a coin-flip test is the market's upward drift, not timing skill (it reads $p = 0.978$ and $p = 1.000$ against the up-day base rate), and it does not survive transaction costs. This matches the efficient-market hypothesis, and the value of the project is that it measures the ceiling carefully instead of claiming to break it. One caveat on the tests themselves: the Diebold-Mariano statistic here assumes serially uncorrelated loss differences, which is optimistic for daily data, so the two significant-worse flags are the ones most sensitive to it. The conclusion does not depend on them.

The variance of returns is predictable where the direction is not, and M3 already put a number on it: the GJR-GARCH variance forecasts score a real out-of-sample QLIKE and a positive Mincer-Zarnowitz R squared (M3.md section 4), unlike any of the mean forecasts here. That points to where a future project could add value, in volatility and risk forecasting rather than return prediction.